

Project Proposal

Audio Emotion Extraction

Speech Emotion Recognition from Audio Signals

AI-Based Programming Project

Team Members

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Project Overview

Audio Emotion Extraction is an end-to-end machine learning system that identifies human emotions from speech recordings. The project uses acoustic signal processing, feature engineering, feature selection, supervised learning, ensemble modeling, and a reasoning layer to produce accurate and interpretable emotion predictions.

The target dataset is CREMA-D, a real speech-emotion dataset containing labeled audio clips across six emotion classes: Angry, Disgust, Fear, Happy, Neutral, and Sad.

Problem Statement

Speech carries emotional information through tone, pitch, energy, rhythm, and spectral patterns. Detecting these signals automatically can support applications such as virtual assistants, user-experience analysis, mental-health screening support, call-center analytics, and human-computer interaction.

The challenge is to convert raw audio into reliable model-ready features, train robust classifiers, and explain why a model produced a specific emotion label.

Objectives

- Build a reproducible audio preprocessing and feature-extraction pipeline.
- Extract meaningful acoustic features such as MFCCs, chroma, spectral statistics, RMS energy, zero-crossing rate, and pitch.
- Apply feature selection using greedy heuristic search and a genetic algorithm.

- Train and compare baseline, selected-feature, voting ensemble, and stacking ensemble models.
- Add a symbolic reasoning layer that explains model outputs using explicit rules.

Methodology

Data Pipeline	Load CREMA-D audio files, parse labels, validate files, trim silence, resample audio, and extract fixed-length feature vectors.
Feature Processing	Handle outliers and missing values, then apply leakage-safe scaling using training data only.
Feature Selection	Use greedy forward selection and genetic algorithm search to reduce redundant or weak acoustic features.
Modeling	Train SVM, KNN, Random Forest, voting ensemble, and stacking ensemble models.
Evaluation	Report accuracy, macro precision, macro recall, macro F1, weighted F1, confusion matrices, and error analysis.
Explainability	Use symbolic rules over model agreement, class reliability, confusion behavior, and acoustic evidence to generate readable explanations.

Expected Outcomes

- A complete notebook-based speech-emotion recognition pipeline.
- Comparative evaluation of baseline, feature-selected, and ensemble models.
- Visual and tabular performance reports.
- A logic-based assistant that gives confidence states, rules fired, and explanation examples.
- A cleaner and more interpretable audio emotion classification workflow.

Tools and Technologies

Language	Python
Audio Processing	Librosa
Machine Learning	Scikit-learn
Optimization	Greedy Feature Selection, Genetic Algorithm
Visualization	Matplotlib, Seaborn
Environment	Jupyter Notebook / Google Colab

Project Scope

The project focuses on supervised emotion classification from speech audio. It does not attempt real-time deployment, speaker identification, or clinical diagnosis. The final system is designed as a research and educational pipeline that emphasizes reproducibility, evaluation quality, and interpretability.

Prepared for the Audio Emotion Extraction notebook project.